



Application of Chlorogenic acid as a substitute for antibiotics in Multidrug-resistant *Escherichia coli*-induced mastitis

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ABSTRACT

Mastitis affects animal welfare and causes economic losses in the dairy industry. It is caused mainly by bacterial pathogens, among which *Escherichia coli* (*E. coli*) is one of the prominent causative agents. To treat bovine mastitis, antibiotics were widely used. However, their extensive and uncontrolled use has led to the emergence of multi-antibiotic-resistant strains. Indeed, a superbug of *E. coli* was successfully isolated from a mastitis-suffering cow and found resistant to at least 10 antibiotics. Therefore, the development of a universal therapeutic agent used as a replacement for the antibiotic is an immediate need in the dairy industry. To do so, we examined whether chlorogenic acid (CGA), a natural and herbal extract, could be a perfect alternative in mastitis treatment. In this study, we observed that the combination of CGA and antibiotic had an additive or synergistic effect; CGA fought against the superbug by directly targeting bacterial cell wall and membrane; CGA can significantly alleviate the mastitis caused by the superbug *E. coli* via its antimicrobial, antioxidant and anti-inflammatory activities. Collectively, these data indicated that CGA had a true potential to replace antibiotics during mastitis treatment.

1. Introduction

Mastitis is a disease highly affecting animal welfare and economically incurring significant losses in the dairy industry. Even though various advanced management practices have been applied in cow rearing, mastitis is still a daunting disease and a severe economic problem worldwide. It not only leads to an increase in the cost of rearing but also adversely impacts the yield and quality of milk. Moreover, mastitis-affected milk may be harmful to humans, because multidrug-resistant pathogens may be transmitted by contaminated unpasteurized milk and antibiotic residues may be contained in milk [1]. Therefore, mastitis is considered a major public health concern as well.

Bovine mastitis including clinical and subclinical forms is the inflammation of the mammary gland/udder that is caused commonly and mainly by several bacterial pathogens [2]. These common etiologic agents are generally categorized as contagious and environmental pathogens. Among these pathogens, *Staphylococcus aureus* (*S. aureus*),

Streptococcus agalactiae (*S. agalactiae*), and *Streptococcus dysgalactiae* (*S. dysgalactiae*) are considered as contagious pathogens, which are able to be spread between cows. *Streptococcus uberis* (*S. uberis*), *Escherichia coli* (*E. coli*), and *Klebsiella pneumoniae* (*K. pneumoniae*) are considered as environmental pathogens, which are opportunistic pathogens and can be transferred from the contaminated environment to the mammary gland of a cow. The complexity of etiological agents challenges the treatment of mastitis. To combat these pathogens and treat mastitis, antibiotic therapy was broadly applied, following the early and rapid detection of etiological agents. However, extensive and uncontrolled use of antibiotics has led to an outbreak of multiple antibiotic-resistant bacterial strains, which make it way more difficult to treat mastitis. The reality is that fewer and fewer antibiotics are effective against these causative pathogens, particularly in Gram-negative bacteria such as *E. coli*. Antibiotic resistance has been regarded as a major healthcare crisis of this century.

To tackle the microbial antibiotic resistance, many countries have

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issued related regulations and policies to limit the use of antibiotics in livestock breeding. On this background, finding a substitute for antibiotics that can be used in the treatment of mastitis is of great significance. Due to their easy-to-obtain and large quantity, much attention has been fused on natural and herbal substances at present. Among them, chlorogenic acid (CGA), also known as coffee tannic and 3-caffeoyl-quinic acid, is one of the most essential and common classes of natural products [3]. It is one of the well-known secondary phenolic metabolites produced by plants during aerobic respiration. It can be largely extracted from green coffee [4], but also found in various other plants, such as Chinese herbal medicines [5–7], vegetables [8,9], and fruits [10–12]. Many studies have indicated that CGA has antioxidant [13] and anti-inflammatory [14] features, antidiabetic and anti-obesity properties [15], antibacterial effects [16], and anti-tumor activity [17]. In addition, CGA has been found to modulate lipid and glucose metabolism in both genetically and healthy metabolic-related disorders [18]. Therefore, it is speculated that CGA can help to treat hepatic steatosis, cardiovascular disease, diabetes, and obesity as well [19]. Further research might be needed. Given its remarkable biological effects presented above, we wondered whether this natural compound consistently functions in *E. coli*-induced mastitis. Thus, we analyzed the potential of CGA to take the place of antibiotics in the treatment of mastitis.

2. Materials and methods

2.1. Reagents

Chlorogenic acid (CGA) was purchased from Sangon Biotech (Shanghai) Co. All the antibiotics used in this study were purchased from Aladdin kit Shanghai.

2.2. Bacterial strain and culture condition

A multidrug resistance (MDR) clinical *E. coli* strain (laboratory ref, ZJ41) was isolated from a dairy cow farm in Hangzhou city, Zhejiang province, China. The strain was cultured at 37 °C, 200 rpm in LB medium.

2.3. Susceptibility testing

Minimum inhibitory concentration (MIC) was determined by broth microdilution according to the Clinical and Laboratory Standards Institute broth microdilution method (CLSI) [20] and European Committee on Antimicrobial Susceptibility Testing (EUCAST) [21]. The working stock solution of each CGA and antibiotics listed in Table 1 were in fresh MHB. For the checkerboard assay, each agent was tested in 2-fold dilution based on the concentration recommended in breakpoint tables for interpretation of MICs [21]. Several 96-well U-shaped-bottom microplates were prepared to cover all the different combinations and single compounds. Each well contained 50 μ L of 2 $\times$ the final concentrations of each compound alone and in combination. Every plate had a negative and positive growth control. In the set of MIC determinations, each well was inoculated with 50 μ L of 0.5×10^6 CFU/mL bacterial suspension prepared from 24 h culture grown in an LB plate. Each set included *E. coli* ATCC 25,922 control strain. After 18 to 20 h of incubation, MICs were manually read as the lowest concentration of the reagent that completely inhibited visible growth.

2.4. Scanning electron microscope

Overnight bacterial culture was inoculated in fresh LB broth at a dilution ratio of 1: 100 and incubated at 37 °C with shaking at 200 rpm. When bacterial growth reached the logarithmic phase, a MIC of CGA or saline (control group) was added into the culture, respectively, and continued incubating for 1 hr. After incubation, the pellets from the two cultures were collected by centrifugation at 6000 r/min for 5 min,

Table 1

Minimum inhibitory concentration (MIC, μ g/mL) of antibiotics alone or in combination with chlorogenic acid.

Reagents	MIC (alone)	MIC (combined)		FICI ^a	Effects
		Chlorogenic acid	Antibiotic		
Chlorogenic acid	2000				
Penicillin	512	1000	256	1	Additive
Erythromycin	512	1000	128	0.75	Additive
Sulfisoxazole	2048	1000	512	0.75	Additive
Ampicillin	512	1000	512	1.5	Indifference
Cepharmycin C	512	250	32	0.315	Synergism
Timicosin	512	500	128	0.5	Synergism
Spectinomycin	32	1000	32	2	Indifference
Tobramycin	32	500	16	0.75	Additive
Doxycycline	128	500	64	0.75	Additive
Streptomycin sulfate	512	1000	256	1	Additive
Amoxicillin	512	1000	256	1	Additive

^a Fractional Inhibitory Concentration Index (FICI) = MIC (A in the presence of B) / MIC (A alone) + MIC (B in the presence of A) / MIC (B alone). FICI of <0.5 indicates synergism, >0.5–1 indicates additive effects, >1 to <2 indifference, and ≥ 2 is considered to be antagonism.

washed twice with 2 mL of PBS, and finally resuspended in 1 mL of 2 % pentanol for 8–12 hr of fixation at 4 °C. After fixation, bacterial pellets were rinsed with PBS, and then dehydrated by incubating in 30 % ethanol, 50 % ethanol, 70 % ethanol, 80 % ethanol, and 90 % ethanol by order for 10 min each time. Finally, bacterial pellets were incubated in 100 % ethanol for 10 min twice. Once bacteria were prepared, the morphological and structural changes of bacteria were observed by scanning electron microscope.

2.5. Mouse mastitis model

6–8 week-old C57BL/6 mice were purchased from Shanghai Zoo Multiplying Farm and fed in the laboratory animal center of Nanjing Agriculture University. 5–7 days after giving birth, a 10–14 week-old mouse was injected with 2.5×10^7 CFU of *E. coli* into the mammary gland. Regarding CGA treatment, per kg of the mouse was intraperitoneally injected with 10 mg of CGA twice per day and for 3 days. Regarding cefoxitin treatment, per kg of the mouse was intraperitoneally injected with 1 mg of CGA twice per day and for 3 days.

2.6. Real-time Quantitative PCR (RT-qPCR)

Total RNA was extracted from the tissue of the mastitis mouse. The RNA was converted to cDNA (Takara, Dalian, China, ref# RR047A) by reverse transcription. Using cDNA as a template, RT-qPCR reaction. Was performed in SYBR Green Master Mix (Bio-Red, CA) with Applied Biosystem QuantStudio 5 Real-time PCR detection system. The transcript levels of IL-1 β , IL-6, COX-2, SSA, and GAPDH were detected, among which GAPDH was the housekeeping gene. The fold changes were calculated using the method of the 2 to the power of $-\Delta \Delta Ct$. For example, $\Delta Ct(\text{IL-6 of the test group}) = Ct(\text{IL-6 of the test group}) - Ct(\text{GAPDH of the test group})$, $\Delta Ct(\text{IL-6 of the control group}) = Ct(\text{IL-6 of the control group}) - Ct(\text{GAPDH of the control group})$, $\Delta \Delta Ct = \Delta Ct(\text{IL-6 of the test group}) - \Delta Ct(\text{IL-6 of the control group})$.

2.7. Statistical analysis

All values are expressed as means $\pm$ SEM. The statistical comparisons were performed using a one-way ANOVA test by the Prism software (GraphPad, La Jolla, CA). p values < 0.05 (*p < 0.05, **p < 0.01,) were considered to be statistically significant.

3. Results

3.1. CGA had an additive or synergistic effect with antibiotics

To investigate the potential of CGA to replace antibiotics, minimum inhibitory concentration (MIC) and fractional inhibitory concentration Indexes (FICI) were first determined against a multidrug-resistant (MDR) clinical isolate of *E. coli*, which is originally isolated from mastitis-suffering cow and found resistant to at least 10 antibiotics. As shown in Table 1, in the single administration, 2 mg/mL of CGA produced antimicrobial activities, tested by checkerboard assay. Furthermore, the application of CGA and antibiotics in combination was considerable. In most cases, the combination of CGA and antibiotics sharply decreased their MIC and had an additive effect. In addition, the combination of CGA and cephamycin C or tilmicosin had a synergistic activity with a quite lower MIC.

3.2. Chlorogenic acid fights against *E. coli* by targeting bacterial cell wall and membrane

To understand the mode of action of CGA, we detected the morphological changes of *E. coli* upon treatment of CGA by scanning electron microscopy. Prior to that, we analyzed the growth curve of *E. coli* treated with a series of CGA. As shown in Fig. 1A, CGA had the antimicrobial effect at a MIC that would be used in the following experiment. According to the morphological analysis, the CGA-treated cells apparently became rough, irregular, and collapsed, compared to the untreated cells (Fig. 1B). It indicated that the antimicrobial mechanism of CGA was related to the damage of bacterial cell wall and membrane structures. To provide more evidence for this claim, we then detected the release of alkaline phosphatase that exists between bacterial cell wall and membrane. As shown in Fig. 1C, the level of alkaline phosphatase in the culture was significantly increased over time in a dose-dependent manner, indicating that the cell wall was obviously impaired. In addition, we determined the protein and DNA that might be released into the culture. As expected, the concentration of total protein of the group with CGA was much higher than that of the group without

CGA (Fig. 1D), and the level of DNA in the culture was significantly increased due to CGA treatment (Fig. 1E). These data demonstrated that the bacterial membrane was destroyed by CGA. In addition to the efflux of intracellular components, the influx of extracellular substances also occurred during CGA treatment. This notion was demonstrated by the observation that fluorescein Isothiocyanate (FITC) abundantly accumulated in the cell, determined by flow cytometry (Fig. 1F). Collectively, it suggested that CGA can target bacterial cell wall and membrane structures to enhance their permeability.

3.3. CGA protects against *E. coli*-induced mastitis

As shown above, CGA had remarkable antimicrobial activity in vitro. Therefore, we wondered whether CGA had a consistent effect in vivo. To do so, we constructed a mouse mastitis model by the injection of multidrug-resistant *E. coli* through the duct of the mammary gland. As shown in Fig. 2, the mammary gland of mice infected with *E. coli* showed tissue swelling and congestion, strands of fibrin with cellular debris, and infiltration of inflammatory cells (Fig. 2D), compared to the control group (Fig. 2A). However, CGA or antibiotic treatment significantly reduced these syndromes of mastitis (Fig. 2B&C). It indicated that CGA can effectively treat mastitis caused by *E. coli*.

3.4. CGA alleviates the oxidative stress and inflammation of mammary tissue in *E. coli*-induced mastitis

To further evaluate the effect of CGA treatment, we analyzed the biochemical changes of the mammary tissue in *E. coli*-infected mice following the treatment of CGA or antibiotic. First, we observed that bacterial burden in this tissue was significantly decreased due to CGA or ABX treatment (Fig. 3A). But there is no difference between CGA and ABX group. It indicated that CGA had the same antimicrobial effect in vivo as the antibiotic. Then, we observed that CGA or ABX treatment significantly diminished the enzyme activities of *N*-acetyl- β -glucosaminidase (NAGase), a prominent bovine mastitis inflammatory biomarker [22], compared to the non-treatment group (Fig. 3B). It is compatible with the result obtained in Fig. 2. As oxidative stress is

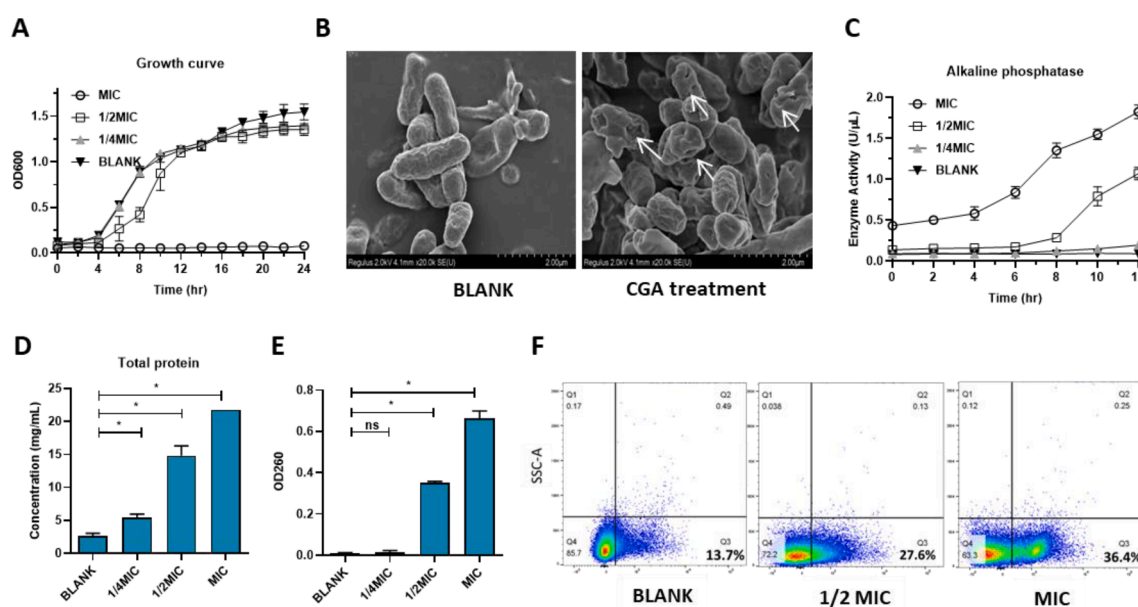


Fig. 1. Chlorogenic acid impairs the cell wall and membrane of *E. coli*. A. Growth curve of *E. coli* treated with a range of concentrations of chlorogenic acid. B. Morphology of *E. coli* upon treatment of chlorogenic acid (GCA) at MIC (2 mg/mL) detected through Scanning Electron Microscope. C. Enzyme activity of alkaline phosphatase in the *E. coli* culture upon treatment with different MIC of GCA. D. Total protein in the *E. coli* culture upon treatment with different MIC of GCA. E. OD260 of the *E. coli* culture upon treatment with different MIC of GCA. F. Fluorescein isoleucyanate assay for *E. coli* upon GCA treatment detected by Flow cytometry. One-way ANOVA (* $p < 0.05$, ** $p < 0.01$).

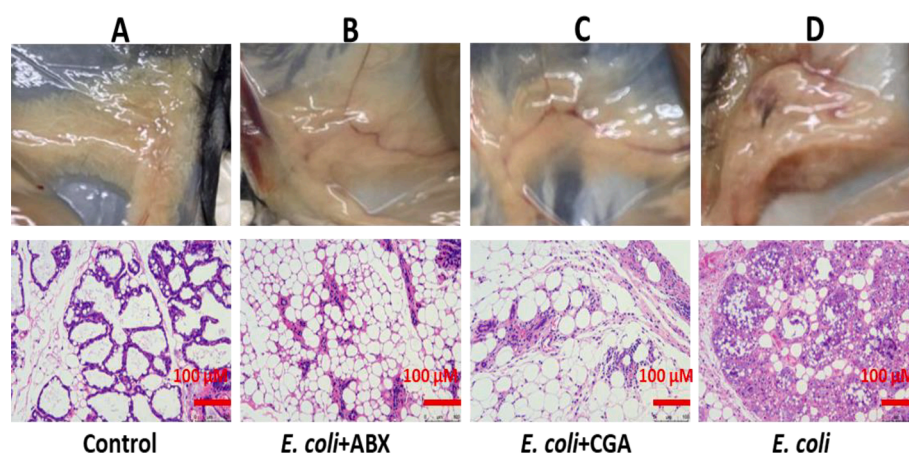


Fig. 2. Direct and histopathological observations of mammary gland by hematoxylin and eosin (H&E) staining. A. Mouse mammary gland was injected with saline as control. B. Mouse with mastitis was treated by intraperitoneal administration of antibiotic cefoxitin (ABX) (1 mg/kg, twice per day and for 3 days). C. Mouse with mastitis was treated by intraperitoneal administration of CGA (10 mg/kg, twice per day and for 3 days). D. Mouse mammary gland was injected with *E. coli* (2.5×10^7 CFU) to construct mastitis.

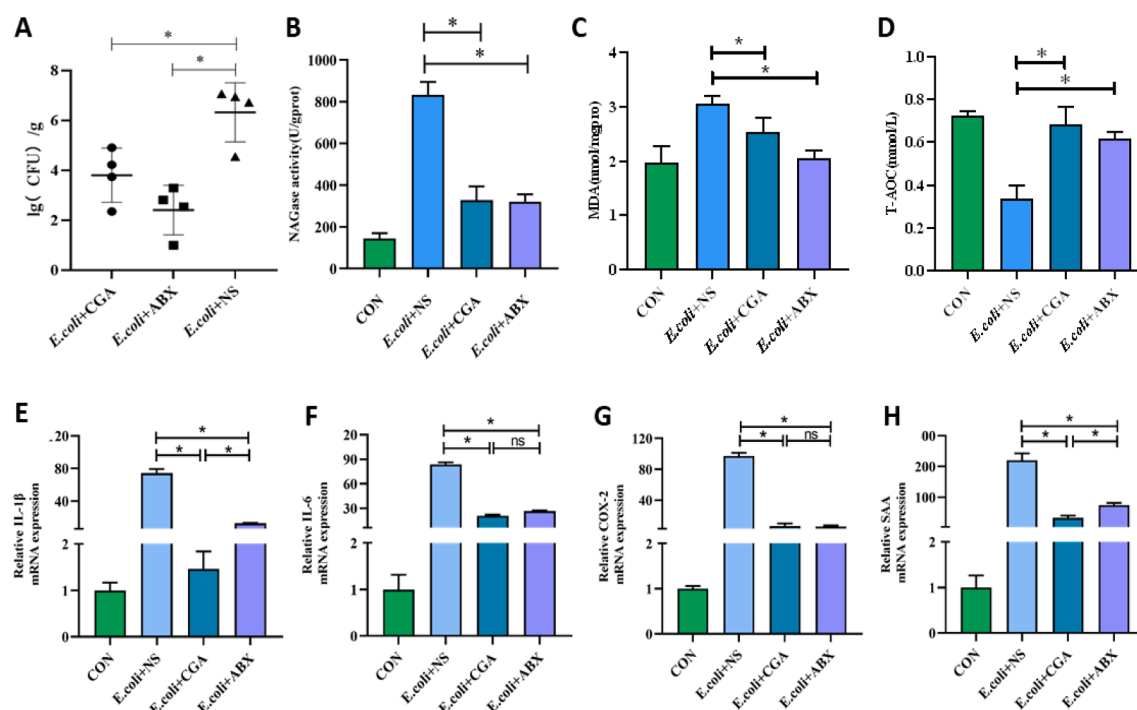


Fig. 3. Chlorogenic acid alleviates *E. coli*-caused oxidative stress and inflammation in the mammary gland. Bacterial burden in the mammary gland (A), *N*-acetyl- β -D-glucosaminidase (NAGase) enzyme activities (B), the malondialdehyde (MDA) release (C), the total antioxidant capacity (T-AOC) release (D), IL-1 β (E), IL-6 (F), COX-2 (G) and SSA (H) were detected in the mammary gland during mastitis. One-way ANOVA (* $p < 0.05$, ** $p < 0.01$).

highly associated with inflammation of the mammary gland in cows [23], we determined the levels of the antioxidant indicators or oxidative stress indicators that reflect oxidative stress in vivo. As expected, The treatment of CGA or antibiotic led to a significant reduction of MDA (Fig. 3C) and a high increase of T-AOC (Fig. 3D), indicating that CGA was able to alleviate the oxidative stress accompanied by mastitis. To evaluate the effect of CGA treatment on inflammation, we detected the regular inflammatory biomarker, IL-1 β , IL-6, COX-2, and SSA, by quantitative real-time PCR. Consistently, the treatment of CGA or antibiotic significantly decreased the expression of these inflammatory mediators (Fig. 3E and F). Moreover, GAC treatment showed a lower level of IL-1 β and SSA than antibiotic treatment. Collectively, it suggested that CGA had significant anti-oxidative and anti-inflammatory effects during mastitis.

3.5. CGA alleviates oxidative stress and inflammation of the liver

E. coli-induced mastitis often accompanies severe liver damage, and the liver plays a key role in the metabolic and inflammatory physiology of the dairy cow in the early lactation which has high metabolic and immunological demands [24]. Therefore, we detected the level of oxidative stress and inflammation of the liver. Consistently, NAGase enzyme activities (Fig. 4A), T-AOC (Fig. 4B), and MDA (Fig. 4C) release, and the inflammatory secretion (Fig. 4D–G) in the liver were impacted by CGA or antibiotic treatment in almost the same manner as those in the mammary gland. It further established that CGA played a significant role in the alleviation of oxidative stress and inflammation.

4. Discussion

Mastitis is considered as a foremost problem in the dairy industry, affecting animal welfare and causing economic losses. The main

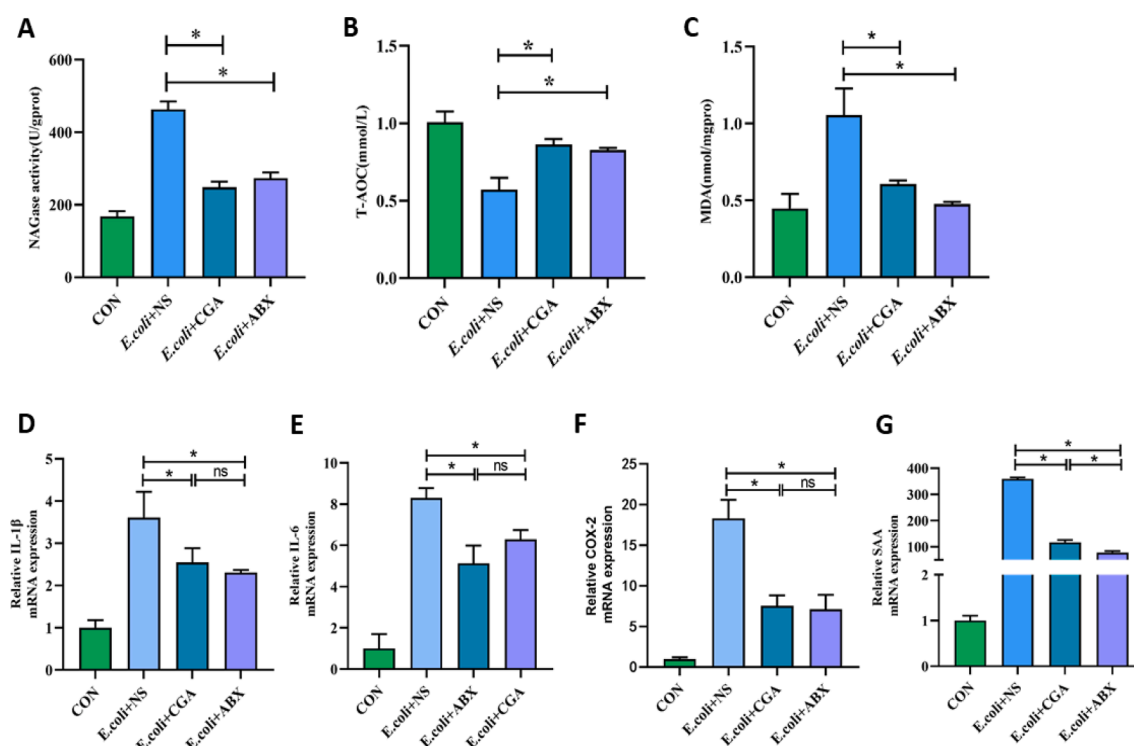


Fig. 4. Chlorogenic acid alleviates oxidative stress and inflammation in the liver. The enzyme activities of NAGase (A), the level of T-AOC (B) MDA (C), and the release of IL-1 β (D), IL-6 (E), COX-2 (F), and SSA (G) were detected in the liver during mastitis. One-way ANOVA (* $p < 0.05$, ** $p < 0.01$).

challenge for this disease in the dairy industry is to treat the sick animal in such a way that it will not compromise the animal's health and the milk quality. Additionally, the diversity of the causative etiological agents makes the treatment even more difficult. In the past decades, antibiotics have been widely used as the sole therapeutic drug for cow mastitis, but it has led to severe antimicrobial resistance and antibiotic residue. Therefore, the development of a universal therapeutic agent used as a replacement for the antibiotic is the immediate need of the dairy industry.

Among the new therapeutic strategies for mastitis, like bacteriocins, probiotics, herbal therapy, immunotherapy, and nanoparticle technology, the herbal extract has been thought to be the most promising candidate [25,26]. The main reason probably is that it is safer to use and easier to obtain than other therapies. More importantly, it can overcome the shortcoming of antibiotics that easily leads to antimicrobial resistance. Therefore, this study focused on the natural plant extract, CGA, and investigated its role in the treatment of mastitis induced by multidrug-resistant *E. coli*. In this study, we well established by electron microscope photography that CGA can directly kill *E. coli* by destroying bacterial cell wall and membrane. It explains the additive or synergistic effect of CGA when combined with antibiotics. The reason probably is that the permeability of bacterial cell wall and membrane structures enhanced by CGA can improve the entrance of antibiotics through the bacterial cell wall, and finally lead to increased sensitivity of antibiotics.

In addition, from some point, the direct antimicrobial activity may also explain why CGA can significantly alleviate the mastitis caused by multidrug-resistant *E. coli*. During the mastitis, CGA significantly reduced bacterial burden in the mammary gland, which indicated that the decrease of bacterial burden could be responsible for less inflammation and oxidative stress. However, we still cannot exclude the direct anti-inflammatory and anti-antioxidant effects of CGA. Based on the result that CGA treatment displayed lower expression of IL-1 β (Fig. 3.E) and SAA (Fig. 3.H) than antibiotic treatment, it suggests that CGA may have a direct anti-inflammatory effect. In addition, it has been reported that increasing bacteria lysis would exacerbate inflammation [27,28].

Considering this, CGA should directly resist inflammation. More evidence will be needed. When it comes to the mechanism of CGA to limit inflammation, known from the LPS-induced mice mastitis [29], CGA may function via suppressing the TLR4-mediated NF- κ B signaling pathway. But, in terms of the mastitis caused by *S. aureus*, the anti-inflammatory effect of CGA may be via suppressing TLR2 [30]. In *E. coli*-induced mastitis, the mechanism of action of CGA remains unclear. TLR4 may also contribute to it. Further study will be needed.

In conclusion, CGA has a direct antimicrobial activity by targeting bacterial cell wall and membrane, and is capable to resist inflammation during mastitis induced by multidrug-resistant *E. coli*. These data indicate that CGA has the true potential to replace antibiotics.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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